1.) (Similar to HW1#1.)

Base case: For $n=1$, we have $1 = 2\cdot 1^2 - 1$, which is true.

Inductive step: Assume

$$1+5+\ldots+(4n-3) = 2n^2-n$$

for some $n \geq 1$. Adding $(4n+1)$ to both sides,

$$\begin{aligned} 1+5+\ldots+(4n-3)+(4n+1) &= 2n^2-n+(4n+1) \\ &= 2(n^2+2n+1)-n-1 \\ &= 2(n+1)^2-(n+1). \end{aligned}$$

So the formula holds for $n+1$.

By induction, we conclude the formula is true $\forall n \geq 1$. $\square$

2.) (Similar to HW1#3b)

(a.) <u>Prove</u>: Fix $Y \subseteq B$ and $y \in f(f^{-1}(Y))$. Then $\exists\, x \in f^{-1}(Y)$ s.t. $f(x) = y$. As $x \in f^{-1}(Y)$, then $f(x) \in Y$, and so $y = f(x) \in Y$.

(b.) <u>Disprove</u>: Consider $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ and $Y = \{-1, 1\}$. Then

$$f^{-1}(\{-1,1\}) = \{-1,1\}, \qquad f(\{-1,1\}) = \{1\}$$

and so

$$f(f^{-1}(\{-1,1\})) = f(\{-1,1\}) = \{1\} \not\subseteq \{-1,1\}$$

(By HW1#3b, we just need to pick a function f that is not surjective.)

3.) (This is HW3#6a.)

As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, $\sup A$, $\inf B \in \mathbb{R}$ exist.

Let $b \in B$. As $a \leq b$ $\forall a \in A$, then b is an upper bound for A, and so

$$\sup A \leq b.$$

This is true $\forall b \in B$. So $\sup A$ is a lower bound for B, and therefore

$$\sup A \leq \inf B. \qquad \square$$

4.) (This is HW4#5.)

Fix $\varepsilon > 0$. As $\lim_{n\to\infty} a_n = a$ and $a > 0$, then $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \sqrt{a} \cdot \varepsilon \qquad \forall n \geq N.$$

Then for $n \geq N$ we have

$$|\sqrt{a_n} - \sqrt{a}| = \left| \frac{(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})}{\sqrt{a_n} + \sqrt{a}} \right| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}}$$

$$\underset{\sqrt{a_n} > 0}{<} \frac{|a_n - a|}{\sqrt{a}} < \frac{\sqrt{a} \cdot \varepsilon}{\sqrt{a}} = \varepsilon$$

So $\lim_{n\to\infty} \sqrt{a_n} = \sqrt{a}$. $\square$